

Fig. 4: Component layout for the PCB



Fig. 5: HyPerterminal window for name and icon



Fig. 6: HyPerterminal window for COM port selection

to T_{OUT} (pin 7) and R_{IN} (pin 8) of the MAX232, respectively. T_{IN} (pin 10) and R_{OUT} (pin 9) of IC2 are connected to pins 2 and 3 of the COM port connector, respectively. The signals provided on these pins are TTL-level and must be boosted and inverted through a MAX232 converter to comply with the RS-232 standard.

The MAX232 has two internal charge-pumps that convert +5V into $\pm 10V$ (unloaded) for RS-232 driver operation. The first converter uses capacitor C7 to double the +5V input to +10V on capacitor C9 at pin 2. The second converter uses capacitor C8 to

invert +10V to -10V on capacitor C10 at pin 6.

All the serial communication is controlled through special function register SCON. This register contains mode-selection bits, ninth data bit for transmit and receive (TB8 and RB8), and serial-port interrupt bits (TI and RI). Serial communication requires standard baud rate. Timer 1 is configured in the auto-reload mode to generate the baud rate. The baud rate is determined as:

$$\text{Baud rate} = (2^{\text{SMOD}} \times \text{Oscillator frequency}) / (32 \times 12 \times (256 - \text{TH1}))$$



Fig. 7: HyPerterminal window for selection of baud rate, data bits, parity, stop bits and flow control

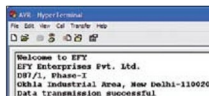


Fig. 8: Message on HyPerterminal screen

To derive the power supply, the 230V AC mains is stepped down by transformer X1 to deliver the secondary output of 9V, 300 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C1 and then regulated by IC 7805 (IC3). Capacitor C2 bypasses the ripples present in the regulated 5V power supply. LED1 acts as the power-on indicator and resistor R1 limits the current through LED1.

Construction and testing

An actual-size, single-side PCB for microcontroller-based serial data transfer circuit is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the PCB to minimise

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